HEDIEH ABDOLLAHI

hedieh.abdollahi@csfk.org — abdollahi.hedieh2013@gmail.com

RESEARCH AFFILIATIONS

Affiliated Researcher, Konkoly Observatory, Budapest, Hungary — Mar 2026 – Present

Visiting Research Fellow, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran — Mar 2026 – Present

POSTDOCTORAL POSITIONS

Postdoctoral Research Fellow, Konkoly Observatory — 1 Nov 2024 – 28 Feb 2026
Budapest, Hungary

Postdoctoral Researcher, Institute for Research in Fundamental Sciences (IPM) — 22 May 2023 – 30 Oct 2024
Tehran, Iran

EDUCATION

Ph.D. in Astronomy, Al-Zahra University, Tehran, Iran — 2016 – 16 Mar 2023
(with two years of research break—officially granted)

Master in Physics, Al-Zahra University, Tehran, Iran — 2013–2015

Bachelor of Physics, Shahed University, Tehran, Iran — 2007–2011

PUBLICATIONS

Peer-Reviewed Journal Articles:

- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies VIII, A Census of Long-Period Variable Stars across the Andromeda Dwarf Satellite System, DOI: 10.3847/1538-4357/ae40f3, *ApJ*
- Discovery and chemo-dynamical properties of two s-process enhanced RR-Lyrae stars, DOI: 10.1051/0004-6361/202557188, *A&A*
- Long-period variable stars in NGC 147 and NGC 185 – II. Their dust production, DOI: 10.3847/1538-4357/adfa2b, *ApJ*
- Search for binary-channel metal-rich RR-Lyrae candidates, DOI: 10.1051/0004-6361/202553790, *A&A*
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies. VI. The Star Formation History and Dust Production in Andromeda IX, DOI: 10.3847/1538-4357/acbbc9, *ApJ*

Manuscripts in Preparation:

- Investigating the Star Formation History and Dust Production of NGC 6822, (In referee review)
- The UKIRT M33 Monitoring Project. VII. A 15-Year K-band Photometric Catalogue of Long-Period Variable Stars in the Central Kiloparsec of M33 (In referee review)
- SEDust: A DUSTY-Based SED-Fitting Pipeline for Dust and Mass-Loss from AGB and LPV Stars (In referee review)
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies IX: Star Formation History and Dust Production in Andromeda Dwarf Galaxies Traced by Long-Period Variable Stars
- Characterizing Time-Resolved RRab Light-Curves to Reveal Additional Modes and Phase Variations
- Recovering the Star Formation History of M31 Using Long-Period Variable Stars

Refereed Conference Proceedings:

- A Deep Dive into Stellar Populations in M33's Central Region: Near-Infrared Observations and Analysis, DOI: 10.52526/25792776-24.71.2-389, BAO
- Half-Light Radius Measurements of Andromeda Dwarf Satellites from the Isaac Newton Telescope Survey Using Exponential, Plummer, and Sérsic Fits, ADS, BAO
- Star Formation History of the Local Group Dwarf Irregular Galaxy, NGC 6822, 2024CoBAO..71..394K, ADS
- Detection of the Long Period Variable Stars of And II Dwarf Satellite galaxy, 2024CoBAO..71..383A, ADS
- Investigating the Dust Production in the Local Group Dwarf Irregular Galaxy, NGC 6822, Through Dusty Stellar Populations, 2024IAUGA..32P1603K, ADS
- Investigating the Evolutionary History of the Local Group Dwarf Irregular Galaxy, NGC 6822, Through Long-Period Variable Stars, 2024IAUGA..32P.648K, ADS
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies VIII, The Star Formation History and Dust Production Rate of Andromeda Dwarf Satellites, 2024eas..conf...41A, ADS
- Comprehensive Compilation of Stellar Data in M33's Central Region: Unveiling the Final-Stage Evolution Through Near-Infrared Survey, 2024eas..conf..826A, ADS
- Thorough Collection of Stellar Information in M33's Core: Revealing Late-Stage Evolution with Near-Infrared Study, 2024eas..conf..912A, ADS
- The formation and evolution of Andromeda IX, 2023arXiv230207599A, ADS
- The role of LPVs in And IX, from star formation to dust production, 2023dewi.confE...8A, ADS
- The Star Formation History and Dust Production in Andromeda IX, Proceedings of the International Astronomical Union, DOI: 10.1017/S1743921323000339, ADS, Cambridge Journal
- INT monitoring survey of Local Group dwarf galaxies: star formation history and chemical enrichment, 2021arXiv210110874P, ADS

Other Scholarly Contributions:

- Physics workbook for high school students, January 2010 (Scientific Editor)

CONFERENCES, SCHOOLS, AND PRESENTATIONS

- 5th Edition of the *Why Galaxies Care About AGB Stars* Conference Series: *3D Winds in the Cosmic Matter Cycle*, Santiago, Chile, 17–21 November 2025 — Contributed Talk
- 403rd IAU Symposium, *The Hidden Beauty of the Galactic Outskirts*, Córdoba, Spain, 6–10 October 2025
- Astronomical Surveys and Big Data 3 (ASBD3), Byurakan, Armenia, 15–19 September 2025 — Contributed Talk
- I-HOW JWST Hands-On Workshop for Central and Eastern Europe, Toruń, Poland, 14–25 July 2025
- EAS Annual Meeting, Cork, Ireland, 23–27 June 2025 — Poster, Zenodo
- 6th Scientific Writing for Young Astronomers, Sintra, Portugal, 5–9 May 2025
- IAU South West and Central Asian Regional Office of Astronomy for Development (SWCA ROAD), Byurakan, Armenia, 16–20 September 2024 — Contributed Talk
- 9th Byurakan International Summer School for Young Astronomers (9BISS), Byurakan, Armenia, 9–13 September 2024
- EAS Annual Meeting, Padova, Italy, 1–5 July 2024 — Poster
- 22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun, San Diego, USA, 24–28 June 2024 — Poster

- 31st Physics Spring Conference (IPM), Tehran, Iran, 16 May 2024 — Poster
- *What Was That?* Planning ESO Follow-up of Transients, Variables, and Solar System Objects in the LSST Era, Garching, Germany, 22–26 January 2024 — Poster, Zenodo 1, Zenodo 2
- *A Decade of ESO Wide-Field Imaging Surveys*, Garching, Germany, 16–21 October 2023 — Contributed Talk, Video, Zenodo
- *Stars (Across the Universe)*, Napoli, Italy, 16–20 October 2023 — Poster, Zenodo
- Asia-Pacific Regional IAU Meeting (APRIM), Koriyama, Japan, 7–11 August 2023 — Poster
- 376th IAU Symposium, Budapest, Hungary, 17–21 April 2023 — Poster
- International Spring School, *Modern Methods of Cosmic Distance Determination*, Konkoly Observatory, Budapest, Hungary, 12–15 April 2023
- XXXI General Assembly of the IAU (IAUGA 2022), IAUS 373, Busan, Republic of Korea, 2–11 August 2022 — Contributed Talk
- 5th INO Workshop on Transient Events and Multi-Messenger Astrophysics, Tehran, Iran, 28–29 July 2022
- 4th INO Workshop on Formation, Evolution, and Asteroseismology of Stars with a View to Exoplanets, Tehran, Iran, 19–20 May 2022
- Iranian National Observatory: 3.4 m Telescope Instrumentation and Science Cases, Tehran, Iran, 2–3 February 2022
- 3rd INO Workshop on Exoplanets and Their Identification, Tehran, Iran, 13–14 January 2022
- 2nd INO Workshop on Star Clusters and the Initial Mass Function, Tehran, Iran, 16–17 December 2021
- 1st INO Workshop on Formation, Evolution, and Structure of Galaxies, Tehran, Iran, 14–15 October 2021
- 2nd Regional Astronomical Summer School on Observational and Data Astronomy and Space Sciences, Byurakan, Armenia, 13–17 September 2021
- Workshop *From Stars to Galaxies: Introduction to Modern Spectroscopic Methods*, Al-Zahra University, Tehran, Iran, 4 January 2020
- Tehran Meeting on Cosmology (IPM), Tehran, Iran, 5–10 August 2017

SOFTWARE TOOLS

Photometry & image processing:	DAOPHOT/ALLSTAR, IRAF, Photutils, DS9, AstroImageJ
Time-series analysis:	Period04, astropy.timeseries (Lomb–Scargle), Lightkurve
SED modeling & radiative transfer:	DUSTY, DESK, *SEDust*
Astronomical survey tools:	TOPCAT, Aladin, Gaia Archive / TAP(+), VizieR, IRSA, SkyMapper, MAST
Database query & automation:	Astroquery, PyVO, Simbad
Version control & environment:	Git, GitHub, Conda, Docker
Operating systems:	Unix/Linux (Ubuntu, CentOS), macOS
Python scientific stack:	Python (NumPy, SciPy, Astropy, Matplotlib, Pandas, Seaborn, IPython, JupyterLab)

ACHIEVEMENTS (FUNDS, GRANTS, AND PRIZES)

- Short-term visit of the ESO, Santiago, Chile, 11–16 November 2025
- The annual Dr. Samimi Prize for the best PhD thesis in astrophysics and cosmology in Iran, 2024
- Short-term visit to the Konkoly Observatory (HUN-REN CSFK), Budapest, Hungary, March–April 2024

RESEARCH INTERESTS

Late stages of stellar evolution; evolved stars (AGB, RGB, LPVs); stellar variability and time-domain astrophysics; stellar winds, mass loss, dust production, and circumstellar environments; radiative transfer and SED modelling; binary evolution; resolved stellar populations and dwarf galaxies; computational astrophysics, scientific software development, and large astronomical surveys.

SUPERVISION / MENTOR

- Ph.D. Thesis Advisor: Unveiling Long-Period Variables in M33's Central Region: Insights into Stellar Evolution and Star Formation via Near-Infrared Photometry
- Project Mentor: A Study on the Evolution of NGC 6822. I. Investigating the Star Formation History Through Evolved Stellar Population

REFERENCES

- Dr. Jacco Th. van Loon

  Lennard-Jones Laboratories, Keele University, ST5 5BG, UK, Email: j.t.van.loon@keele.ac.uk

- Dr. László Molnár

  Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences, MTA Centre of Excellence, Konkoly-Thege Miklós út 15-17, H-1121, Budapest, Hungary
  ELTE Eötvös Loránd University, Institute of Physics and Astronomy, 1117, Pázmány Péter sétány 1/A, Budapest, Hungary, Email: molnar.laszlo@csfk.org

- Dr. Iain McDonald

  Department of Physical Sciences, The Open University, Walton Hall, Milton Keynes, UK
  Jodrell Bank Centre for Astrophysics, Alan Turing Building, University of Manchester, M13 9PL, UK, Email: iain.mcdonald-2@manchester.ac.uk

- Dr. Atefeh Javadi
  Institute for Research in Fundamental Sciences (IPM), Tehran, Iran, Email: atefehj@gmail.com